

Mobile Architecture

Project: Aarvii CCTV AMC Management System **Phase:** D0 — Project Foundation Documentation **Source of truth:** requirements-freeze-v1.md §18 (Mobile Applications) · §20 (Platform reuse)

Two **Flutter** applications are approved for V1, built on the existing platform **Mobile Foundation** (no duplicate implementation, §20).

1. Customer App

Approved features (freeze §18): Dashboard · AMC · Tickets · Invoices · Notifications · Profile

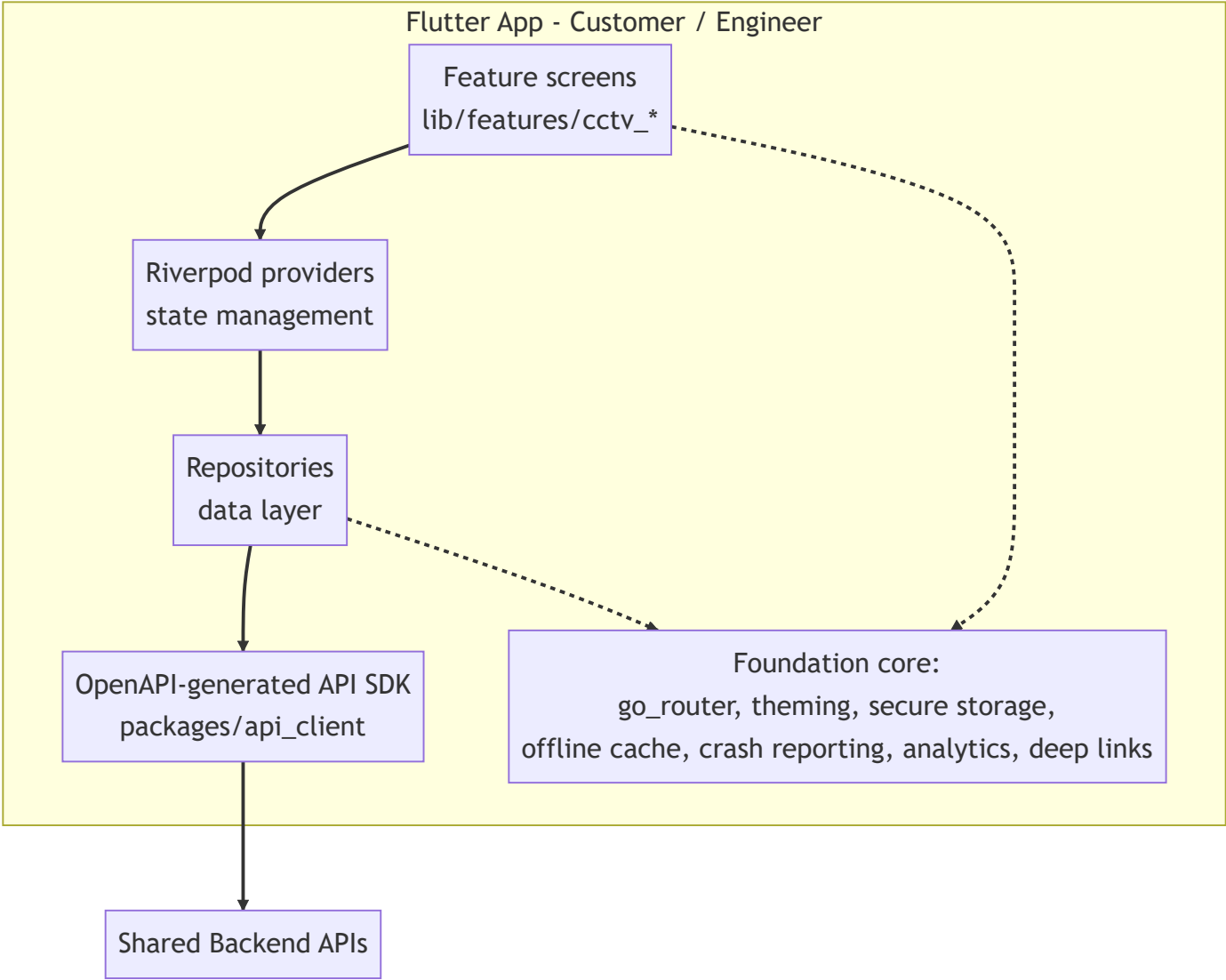
Feature	Behavior (freeze refs)
Dashboard	Summary of the customer's AMC status, upcoming visits, open tickets
AMC	Current active contract term (§8); renewal request (§3)
Tickets	Create, track, reopen closed tickets (§14)
Invoices	View and download invoice PDFs (§16, §19)
Notifications	Receive notification events relevant to the customer (§17)
Profile	Profile management and password reset (OTP, §17)

2. Engineer App

Approved features (freeze §18): Visits · Tickets · Photo Upload · GPS Capture · Signature Capture · **Offline Support**

Feature	Behavior (freeze refs)
Visits	Assigned visit queue; visit execution and report submission (§11–§13)
Tickets	Assigned tickets; create tickets during visits (§14)
Photo Upload	Before/During/After photos; videos (§12, §15)
GPS Capture	Latitude, longitude, timestamp recorded with the visit (§12)
Selfie Capture	Mandatory engineer selfie for completion (§12)
Signature Capture	Customer signs on the engineer's device (§12)
Offline Support	Field operation without connectivity; sync on reconnect (§18)

3. Flutter architecture (inherited from Mobile Foundation)



Concern	Approach (platform ADRs)
State management	Riverpod (ADR-Mobile-0001)
Navigation	go_router (ADR-Mobile-0002)
Environments	dev/qa/uat/prod build flavors (ADR-Mobile-0003)
API client	OpenAPI-generated SDK (ADR-Mobile-0005)
Feature structure	lib/features/<feature>/ — pages, providers, data, models, widgets (business-module convention)
Crash/analytics	Platform abstractions (ADR-Mobile-0008, ADR-Mobile-0009)
Release	fastlane + existing CI pipelines (ADR-Mobile-0010..0012)

CCTV mobile features are added as **new feature slices** consuming the generated SDK — the foundation itself is not modified (freeze \$20).

4. Offline support (Engineer App, \$18)

Built on the foundation's offline cache (ADR-Mobile-0007):

Concern	Design
Read-side	Assigned visits/tickets cached locally for offline viewing
Write-side	Visit evidence (photos, selfie, GPS, signature, remarks) captured and queued locally when offline
Sync	Queued submissions upload when connectivity returns; report submission completes server-side validation (§12 checklist) on sync
Evidence timestamps	GPS timestamp recorded at capture time, not at sync time (§12)
Conflicts	Visit/ticket status transitions validated by the backend on sync; rejected submissions surface to the engineer for correction

Note: the Customer App has no approved offline requirement; offline support is an Engineer App feature per freeze §18.

5. Notifications

Channel	Usage
Email + SMS	The approved business notification channels (§17) — delivered server-side
Push (foundation capability)	The Customer App lists Notifications as a feature (§18); in-app notification display uses the Mobile Foundation's notification/push plumbing (ADR-Mobile-0006) to surface the approved events

Approved events surfaced to mobile users (§17): Ticket Created/Assigned/Closed, Visit Scheduled/Completed, AMC Expiry Reminder, Invoice Generated, OTPs.

6. Authentication

Concern	Design
Login	Platform Auth (JWT/OAuth2) via the foundation auth feature; Login OTP supported (§17)
Token storage	Secure storage (ADR-Mobile-0004)
Password reset	OTP-based (§17)
Roles	Customer App ↔ Customer role; Engineer App ↔ Engineer role; server-side enforcement of §3/§15 permissions
Sessions	Platform session management (list/revoke) available via foundation

App feature ↔ module traceability

App feature	Backend module
Customer: Dashboard / AMC	AMC Contracts (7), Customer Portal (14)
Customer: Tickets	Ticket Management (10)

App feature	Backend module
Customer: Invoices	Invoice Management (12)
Customer: Notifications / Profile	Notifications (§17), Customer Management (3), platform Auth
Engineer: Visits	Service Scheduling (8), Visit Management (9)
Engineer: Tickets	Ticket Management (10)
Engineer: Media / GPS / Signature	Visit Management (9), platform Files

Related documents

- [application-architecture.md](#)
- [navigation-map.md](#)
- [screen-inventory.md](#)
- Platform mobile docs: [../mobile/mobile-platform-manifest.md](#)